Dennis Farmer

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Education

University of Michigan - Ann Arbor, MI **Anticipated May 2027**
B.S.E. Data Science - College of Engineering, Minor in Music
Current: Bayesian Methods for Machine Learning, GIS for Earth Sciences, Generative AI for Music
Coursework: Machine Learning, Regression Analysis, Web Systems, Data Structures and Algorithms, Probability Theory, Linear Algebra, Computational Methods for Statistics, Music & AI

Technical Experience

SuperConductor - Human-AI Music Co-Creativity **December 2025 – Present**
UM ArtsEngine - Professor Herman *Ann Arbor, MI*
- Working with an interdisciplinary team on an interactive transformer-based machine learning model for real-time music generation, controlled via hand gestures and body movements as an art installation

Machine Learning Researcher **May 2025 – September 2025**
Cooperative Institute for Great Lakes Research *Ann Arbor, MI*
- Utilized a Bayesian active learning framework to recommend new buoy sensor locations for improved weather forecasting, leveraging PyTorch and HPC
- Trained convolutional gaussian neural processes on multi-modal environmental Great Lakes data for statistical downscaling and surface temperature uncertainty quantification (PyTorch)

Roaming Behavior of Domestic Cats **October 2024 - December 2024**
Computational Methods for Statistics - Final Research Paper *Ann Arbor, MI*
- Developed a beta regression model based on demographic variables to predict proportion of time domestic cats spent away from home, with negligible bias and MSE verified with Monte Carlo simulations
- Concluded cats roam farther away in dry conditions than during rainfall with a linear mixed-effects model, with bootstrapped 95% confidence interval for difference in means of (0.276 km, 1.069 km)

Projects

Shazam Music Recognition Clone - Project Lead **June 2025 – November 2025**
Michigan Data Science Team *Ann Arbor, MI*
- Designed and taught an 8-week curriculum to teach project members the short-time Fourier transform, the Shazam algorithm, SQL, and React Native, for members to create their own version of Shazam

Sonification of Principal Component Analysis for Music Recommendation **December 2025**
Digital Sound Synthesis - Final Project *Ann Arbor, MI*
- Designed an interpretable content-based music recommender system dashboard, applying principal component analysis to MusiCNN feature embeddings of songs and selecting recommendations based on k-nearest neighbors of songs
- Generated sonifications of songs by mapping principal component axes to synthesizer parameters, with parameterization guided by the correlation loadings between principal components and feature embeddings

Search Engine with MapReduce **November 2025**
Web Systems - Final Project *Ann Arbor, MI*
- Built a scalable full stack search engine for Wikipedia pages using a service-oriented architecture
- Designed and implemented a multi-stage MapReduce pipeline executed with Hadoop for creating a segmented inverted index of 3064 crawled Wikipedia pages
- Created a REST API for distributed search query result retrieval, based on tf-idf text analysis and PageRank

Convolutional Neural Network for Image Classification **September 2023 – December 2023**
Michigan Data Science Team *Ann Arbor, MI*
- Designed a convolutional neural network architecture to distinguish between AI-generated faces and real human faces with a team of five people (PyTorch)
- Wrote a web application for uploading images and receiving classification results via API, with interpretability via GradCam convolutional layer visualizations (React, Express)

Skills

Languages: C++, Python (PyTorch, PySpark), JavaScript (React), HTML, CSS, R, SQL, Bash, ChucK
Tools and Software: Git, Docker, ngrok, Vim, Make, VS Code, GNU Debugger (gdb), Valgrind